At $R(-2, 3)$

$RT^2 = 4 + 9 + 8 + 12 - 8 = 25 > 0$

At $S(2, 2)$

$ST^2 = 4 + 4 - 8 + 8 - 8 = 0$

$\therefore$ The point P lies inside the circle. The points Q and R lie outside the circle and the point S lies on the circle.

Result Condition for the straight line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$ is $c^2 = a^2(1 + m^2)$

Example 3.30

Find the value of k so that the line $3x + 4y - k = 0$ is a tangent to $x^2 + y^2 - 64 = 0$

Solution

The given equations are $x^2 + y^2 - 64 = 0$ and $3x + 4y - k = 0$

The condition for the tangency is $c^2 = a^2(1 + m^2)$

Here $a^2 = 64$, $m = \frac{-3}{4}$ and $c = \frac{k}{4}$

$$c^2 = a^2(1 + m^2) \Rightarrow \frac{k^2}{16} = 64\left(1 + \frac{9}{16}\right)$$

$$k^2 = 64 \times 25$$

$$k = \pm 40$$

Exercise 3.5

1. Find the equation of the tangent to the circle $x^2 + y^2 - 4x + 4y - 8 = 0$ at $(-2, -2)$
2. Determine whether the points $P(1, 0)$, $Q(2, 1)$ and $R(2, 3)$ lie outside the circle, on the circle or inside the circle $x^2 + y^2 - 4x - 6y + 9 = 0$
3. Find the length of the tangent from $(1, 2)$ to the circle $x^2 + y^2 - 2x + 4y + 9 = 0$
4. Find the value of P if the line $3x + 4y - P = 0$ is a tangent to the circle $x^2 + y^2 = 16$

3.5 Conics

Definition 3.3

If a point moves in a plane such that its distance from a fixed point bears a constant ratio to its perpendicular distance from a fixed straight line, then the path described by the moving point is called a conic.

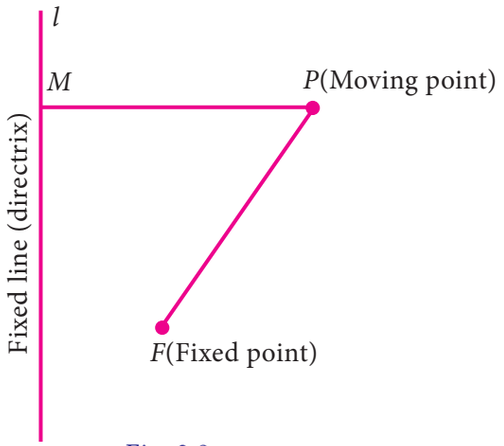


Fig. 3.9

In figure, the fixed point F is called focus, the fixed straight line l is called directrix and P is the moving point such that $\frac{FP}{PM} = e$, a constant. Here the locus of P is called a conic and the constant 'e' is called the eccentricity of the conic.

Based on the value of eccentricity we can classify the conics namely,

a) If $e = 1$, then, the conic is called a parabola

b) If $e < 1$, then, the conic is called an ellipse

c) If $e > 1$, then, the conic is called a hyperbola.

The general second degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents,

(i) a pair of straight lines if $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

(ii) a circle if $a = b$ and $h = 0$

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